

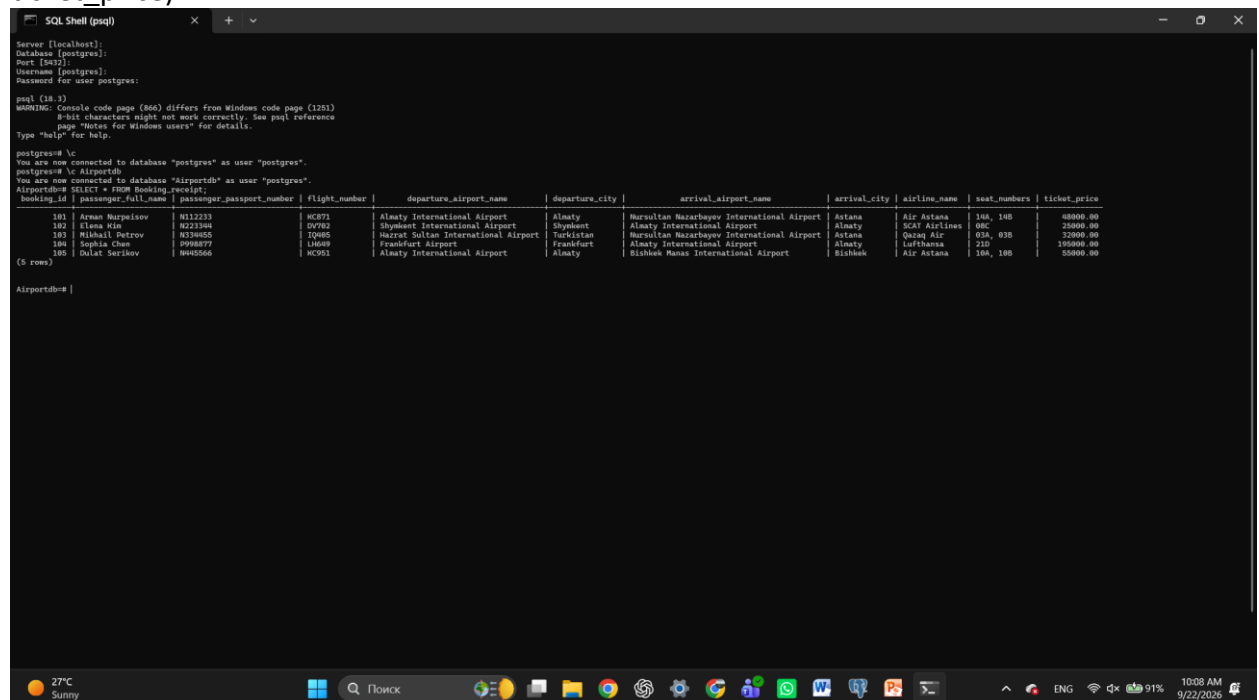
LABORATORY WORK #2

1. Normalization

Task 1

Original Table:

Booking_receipt(
booking_id,
passenger_full_name,
passenger_passport_number,
flight_number,
departure_airport_name,
departure_city,
arrival_airport_name,
arrival_city,
airline_name,
seat_numbers,
ticket_price)



```
SQL Shell (psql)
Server [localhost]:
Database [postgres]:
User [postgres]:
Password for user postgres:

psql (18.3)
WARNING: Console code page (866) differs from Windows code page (1251)
        8-bit characters might not work correctly. See psql reference
        page "Notes for Windows users" for details.
Type "help" for help.

postgres=# \c
You are now connected to database "postgres" as user "postgres".
postgres=# \c Airportdb
You are now connected to database "Airportdb" as user "postgres".
Airportdb=# SELECT * FROM Booking_receipt;
 booking_id | passenger_full_name | passenger_passport_number | flight_number | departure_airport_name | departure_city | arrival_airport_name | arrival_city | airline_name | seat_numbers | ticket_price
-----
101 | Arman Narpisov      | H12233                    | HC871         | Almaty International Airport | Almaty         | Nursultan Nazarbayev International Airport | Astana       | Air Astana   | 10A, 10B    | 40000.00
102 | Elena Kim           | H23304                    | DV782         | Shymkent International Airport | Shymkent       | Almaty International Airport | Almaty       | SCAI Airline | 08C          | 20000.00
103 | Mikhail Petrov      | H33055                    | IQ405         | Hazrat Sultan International Airport | Turkistan      | Nursultan Nazarbayev International Airport | Astana       | Qazaq Air   | 03A, 03B    | 32000.00
104 | Sophia Chen         | H99877                    | U6609         | Frankfurt Airport | Frankfurt      | Almaty International Airport | Almaty       | Lufthansa   | 21D          | 35000.00
105 | Dulat Serikov       | H04566                    | HC951         | Almaty International Airport | Almaty         | Bishkek Manas International Airport | Bishkek      | Air Astana   | 10A, 10B    | 50000.00
(5 rows)
```

The original table violates 1NF because the seat_numbers attribute can contain multiple values in one cell, for example "12A, 12B". In 1NF, each attribute must contain an atomic value

Task 3

To justify the normalization steps, we identify the following functional dependencies (FD):

1. passenger_passport_number → passenger_full_name
2. departure_airport_name → departure_city
3. arrival_airport_name → arrival_city
4. flight_number → airline_name, departure_airport_name, arrival_airport_name
5. booking_id → flight_number, ticket_price
6. (booking_id, passenger_passport_number) → seat_number

Task 2

1NF:

```

SQL Shell (psql)
psql (18.3)
WARNING: Console code page (866) differs from Windows code page (1251)
8-bit characters might not work correctly. See psql reference
page "Notes for Windows users" for details.
Type "help" for help.

postgres=# \c
You are now connected to database "postgres" as user "postgres".
postgres=# \c Airportdb
You are now connected to database "Airportdb" as user "postgres".
Airportdb=# SELECT * FROM Booking_receipt;
 booking_id | passenger_passport_number | flight_number | departure_airport_name | departure_city | arrival_airport_name | arrival_city | airline_name | seat_numbers | ticket_price
-----
101 | Arman Murpeisov | N12233 | Almaty International Airport | Almaty | Nurultan Nazarbayev International Airport | Astana | Air Astana | 10A, 14B | 48000.00
102 | Elena Kim | N22304 | Shymkent International Airport | Shymkent | Almaty International Airport | Almaty | SCAI Airlines | 80C | 25000.00
103 | Mikhail Petrov | N33405 | Nurat Sultan International Airport | Turkistan | Nurultan Nazarbayev International Airport | Astana | Qazaq Air | 03A, 03B | 32000.00
104 | Sophia Chen | P998877 | Frankfurt Airport | Frankfurt | Almaty International Airport | Almaty | Lufthansa | 21D | 195000.00
105 | Dilat Serliov | N45566 | Almaty International Airport | Almaty | Bishkek Manas International Airport | Bishkek | Air Astana | 10A, 10B | 55000.00
(5 rows)

Airportdb=# SELECT * FROM Booking_receipt_1NF;
 booking_id | passenger_passport_number | passenger_full_name | flight_number | departure_airport_name | departure_city | arrival_airport_name | arrival_city | airline_name | seat_number | ticket_price
-----
101 | N12233 | Arman Murpeisov | N12233 | Almaty International Airport | Almaty | Nurultan Nazarbayev International Airport | Astana | Air Astana | 10A | 48000.00
101 | N12233 | Arman Murpeisov | N12233 | Almaty International Airport | Almaty | Nurultan Nazarbayev International Airport | Astana | Air Astana | 14B | 48000.00
102 | N22304 | Elena Kim | N22304 | Shymkent International Airport | Shymkent | Almaty International Airport | Almaty | SCAI Airlines | 80C | 25000.00
103 | N33405 | Mikhail Petrov | N33405 | Nurat Sultan International Airport | Turkistan | Nurultan Nazarbayev International Airport | Astana | Qazaq Air | 03A | 32000.00
104 | P998877 | Sophia Chen | P998877 | Frankfurt Airport | Frankfurt | Almaty International Airport | Almaty | Lufthansa | 21D | 195000.00
(5 rows)

Airportdb=#

```

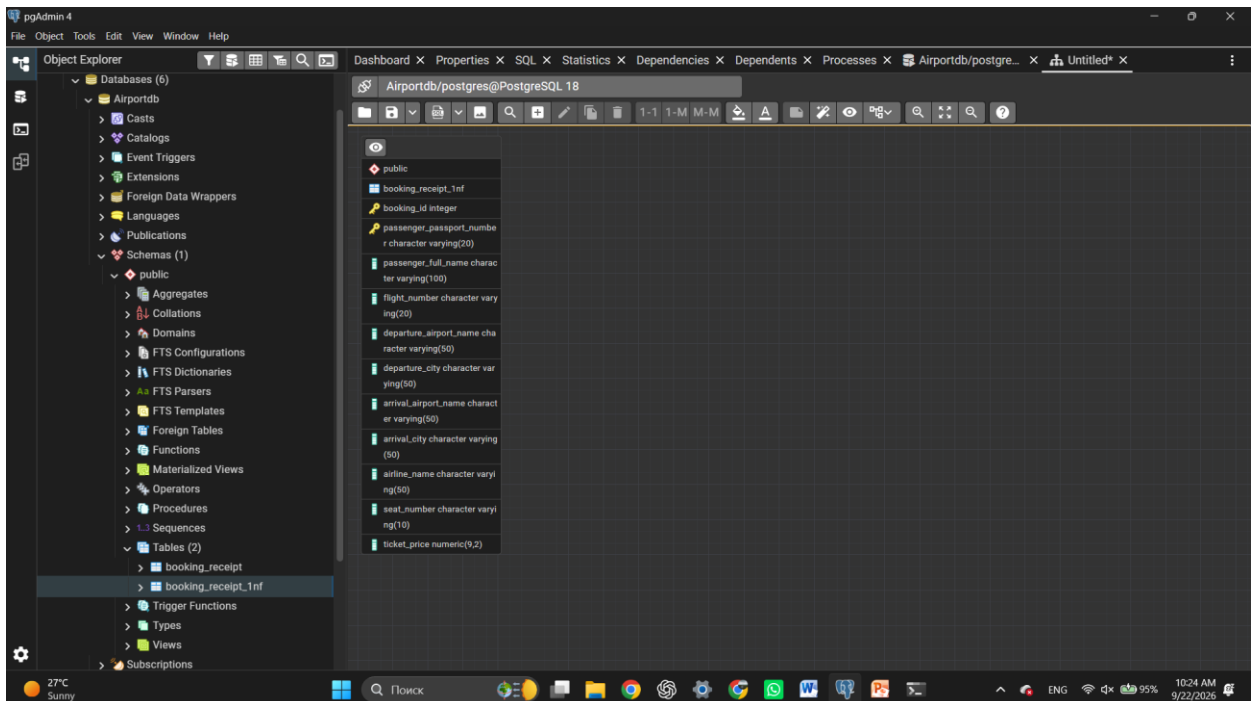
To bring the table to 1NF, I removed the list of seats (seat_numbers) from a single column. Now, each row contains only one atomic seat value (seat_number). Since one booking can have multiple passengers, the primary key is a composite key: (booking_id, passenger_passport_number).

Table: Booking_receipt_1NF

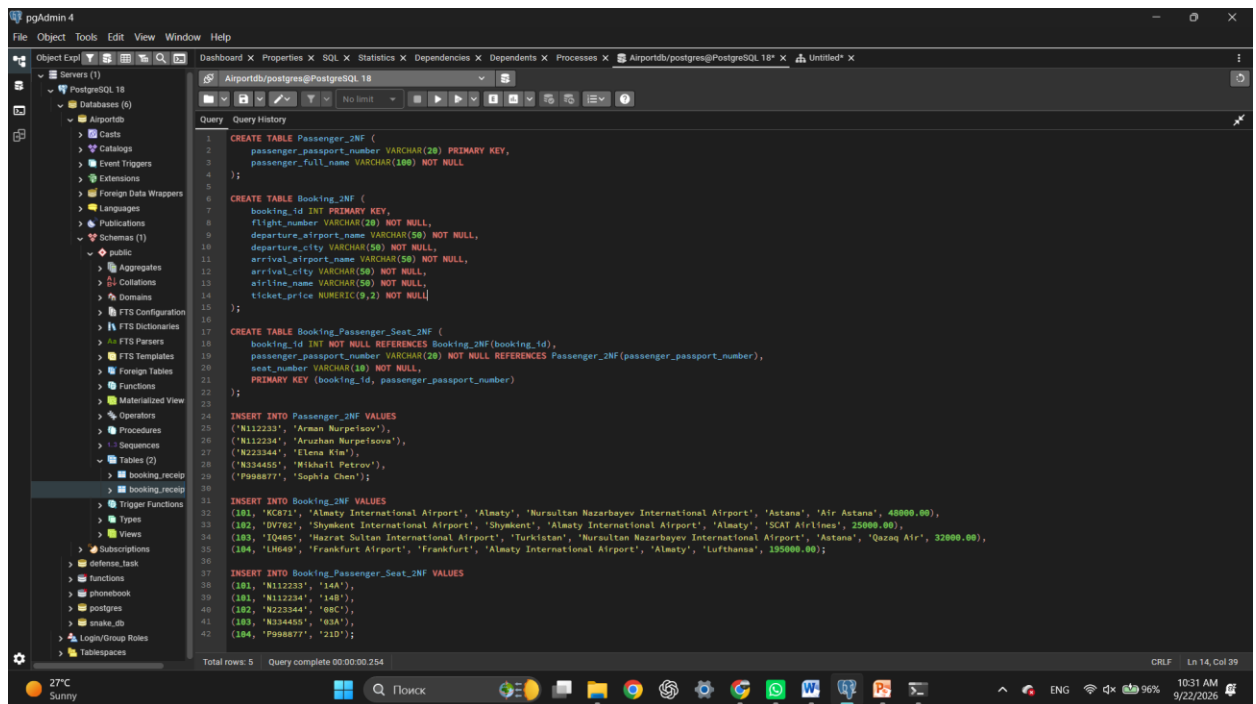
Primary Key (PK): booking_id, passenger_passport_number

Foreign Keys (FK): None

ERD 1NF:



2NF:



2NF: To reach 2NF, I removed partial functional dependencies. Attributes like passenger name depended only on passenger_passport_number, not on the whole primary key. I split the table into three new tables: Passenger_2NF, Booking_2NF, and a junction table Booking_Passenger_Seat_2NF.

Tables and Keys:

1. Passenger_2NF :

Primary Key (PK): passenger_passport_number

2. Booking_2NF :

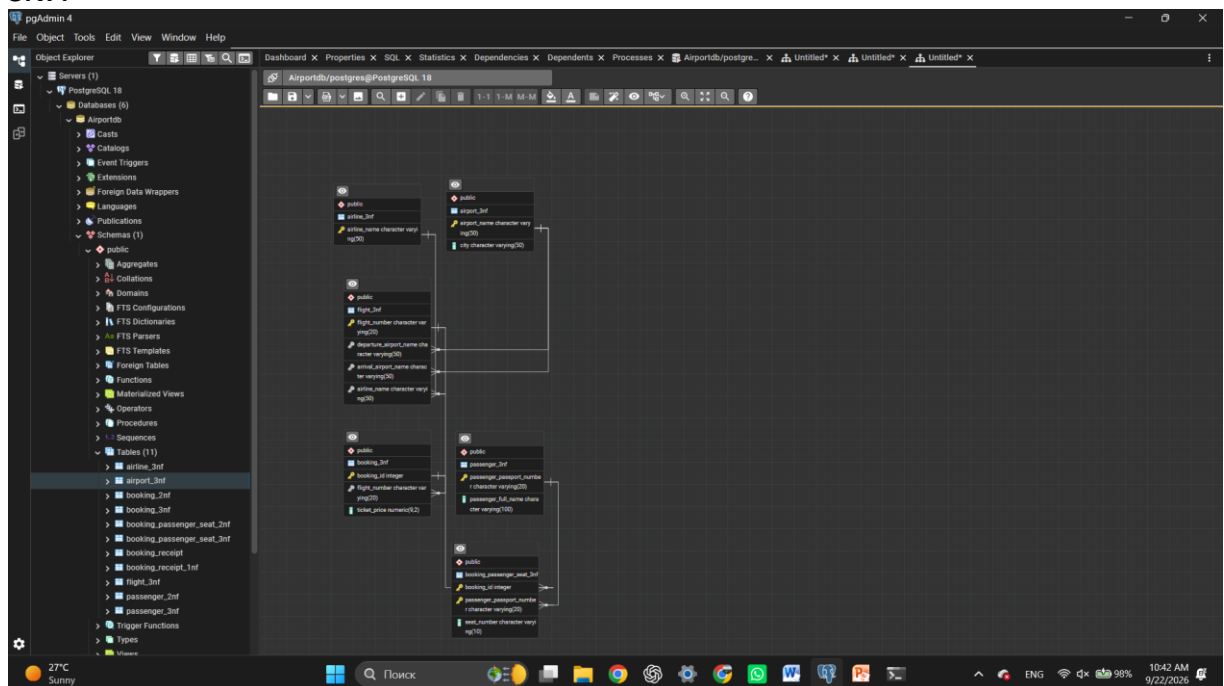
Primary Key (PK): booking_id

3. Booking_Passenger_Seat_2NF :

Primary Key (PK): (booking_id, passenger_passport_number)

Foreign Keys (FK): booking_id → Booking_2NF(booking_id); passenger_passport_number → Passenger_2NF(passenger_passport_number)

3NF:



3NF: To reach 3NF, I removed transitive functional dependencies where non-key attributes depended on other non-key attributes (for example, departure_city and arrival_city depended on airport names, and airline_name depended on flight_number). I created separate tables for Airport_3NF, Airline_3NF, and Flight_3NF.

Tables and Keys:

1.Airport_3NF:

Primary Key (PK): airport_name

Foreign Keys (FK): None

2.Airline_3NF:

Primary Key (PK): airline_name

Foreign Keys (FK): None

3.Flight_3NF:

Primary Key (PK): flight_number

Foreign Keys (FK): departure_airport_name → Airport_3NF(airport_name),

arrival_airport_name → Airport_3NF(airport_name),

airline_name → Airline_3NF(airline_name)

4.Passenger_3NF:

Primary Key (PK): passenger_passport_number

Foreign Keys (FK): None

5.Booking_3NF:

Primary Key (PK): booking_id

Foreign Keys (FK): flight_number → Flight_3NF(flight_number)

6.Booking_Passenger_Seat_3NF:

Primary Key (PK): (booking_id, passenger_passport_number)

Foreign Keys (FK): booking_id → Booking_3NF(booking_id),

passenger_passport_number → Passenger_3NF(passenger_passport_number)

Task 5

I created the 3NF database schema using CREATE TABLE statements with primary and foreign keys. Then, I populated the tables with 5 rows of sample data using INSERT queries

The screenshot displays the pgAdmin 4 interface. The left pane shows the 'Object Explorer' with the 'Airportdb' database selected. The right pane shows the 'Query' editor with the following SQL code:

```
CREATE TABLE Airport_3NF (
  airport_name VARCHAR(50) PRIMARY KEY,
  city VARCHAR(50) NOT NULL
);

CREATE TABLE Airline_3NF (
  airline_name VARCHAR(50) PRIMARY KEY
);

CREATE TABLE Flight_3NF (
  flight_number VARCHAR(20) PRIMARY KEY,
  departure_airport_name VARCHAR(50) NOT NULL REFERENCES Airport_3NF(airport_name),
  arrival_airport_name VARCHAR(50) NOT NULL REFERENCES Airport_3NF(airport_name),
  airline_name VARCHAR(50) NOT NULL REFERENCES Airline_3NF(airline_name)
);

CREATE TABLE Passenger_3NF (
  passenger_passport_number VARCHAR(20) PRIMARY KEY,
  passenger_full_name VARCHAR(100) NOT NULL
);

CREATE TABLE Booking_3NF (
  booking_id INT PRIMARY KEY,
  flight_number VARCHAR(20) NOT NULL REFERENCES Flight_3NF(flight_number),
  ticket_price NUMERIC(9,2) NOT NULL
);

CREATE TABLE Booking_Passenger_Seat_3NF (
  booking_id INT NOT NULL REFERENCES Booking_3NF(booking_id),
  passenger_passport_number VARCHAR(20) NOT NULL REFERENCES Passenger_3NF(passenger_passport_number),
  seat_number VARCHAR(18) NOT NULL,
  PRIMARY KEY (booking_id, passenger_passport_number)
);
```

Below the schema creation, the same query editor shows the insertion of sample data:

```
INSERT INTO Airport_3NF VALUES
('Almaty International Airport', 'Almaty'),
('Nursultan Nazarbayev International Airport', 'Astana'),
('Shymkent International Airport', 'Shymkent'),
('Hazrat Sultan International Airport', 'Turkistan'),
('Frankfurt Airport', 'Frankfurt');

INSERT INTO Airline_3NF VALUES
('Air Astana'),
('SCAT Airlines'),
('Qazaq Air'),
('Lufthansa');

INSERT INTO Flight_3NF VALUES
('KC871', 'Almaty International Airport', 'Nursultan Nazarbayev International Airport', 'Air Astana'),
('DV702', 'Shymkent International Airport', 'Almaty International Airport', 'SCAT Airlines'),
('IQ405', 'Hazrat Sultan International Airport', 'Nursultan Nazarbayev International Airport', 'Qazaq Air'),
('LH649', 'Frankfurt Airport', 'Almaty International Airport', 'Lufthansa');

INSERT INTO Passenger_3NF VALUES
('N112233', 'Arman Nurpeisov'),
('N112234', 'Arzhan Nurpeisova'),
('R223344', 'Elena Kie'),
('N334455', 'Mikhail Petrov'),
('P998877', 'Sophia Chen');

INSERT INTO Booking_3NF VALUES
(101, 'KC871', 48000.00),
(102, 'DV702', 25000.00),
(103, 'IQ405', 32000.00),
(104, 'LH649', 195000.00);

INSERT INTO Booking_Passenger_Seat_3NF VALUES
(101, 'N112233', '14A'),
(102, 'N112234', '14B'),
(102, 'N223344', '8B'),
(103, 'N334455', '83A'),
(104, 'P998877', '21D');
```

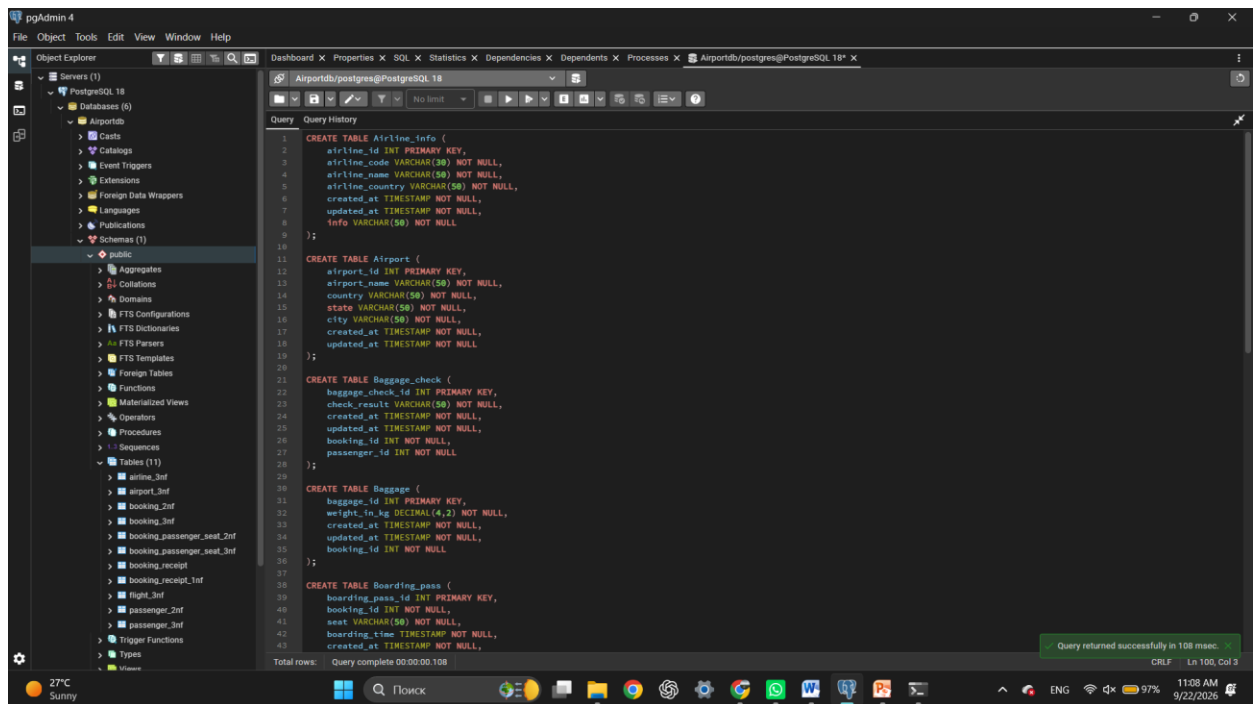
The bottom part of the image shows a terminal window with the following output:

```
Airportdb=# SELECT * FROM Booking_3NF;
 booking_id | flight_number | ticket_price
-----|-----|-----
      101 | KC871        | 48000.00
      102 | DV702        | 25000.00
      103 | IQ405        | 32000.00
      104 | LH649        | 195000.00
(4 rows)

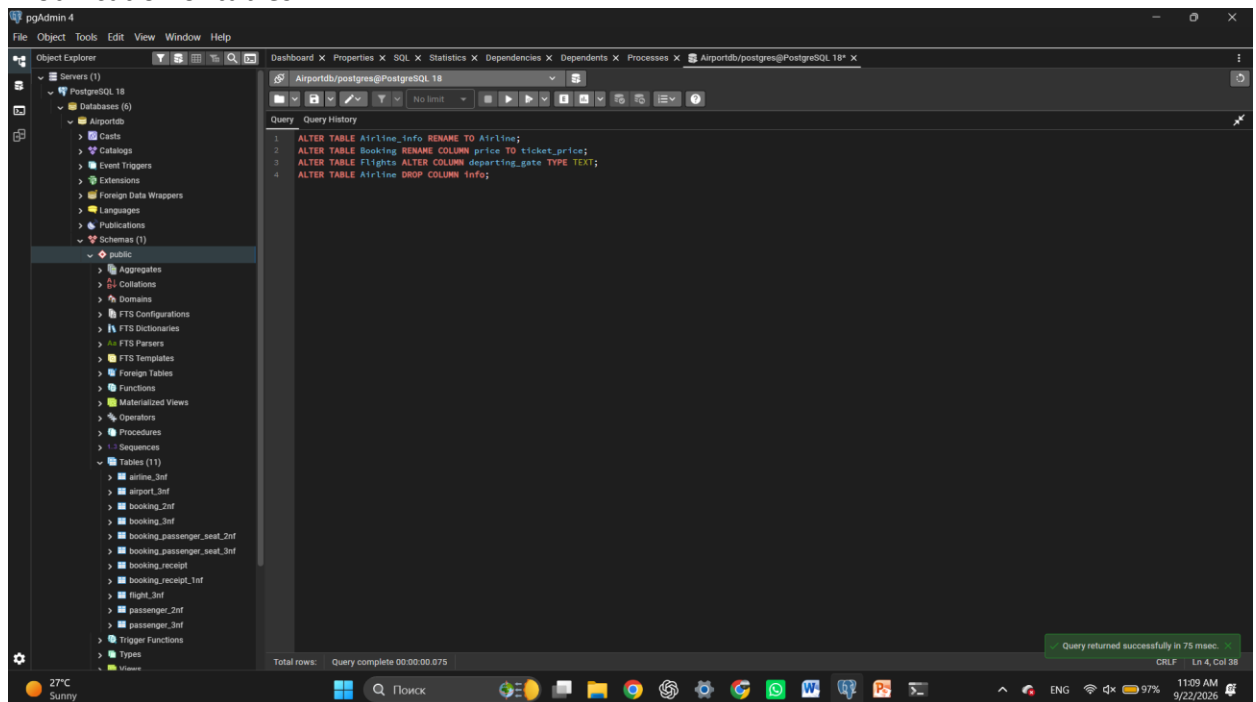
Airportdb=#
```

2. DDL

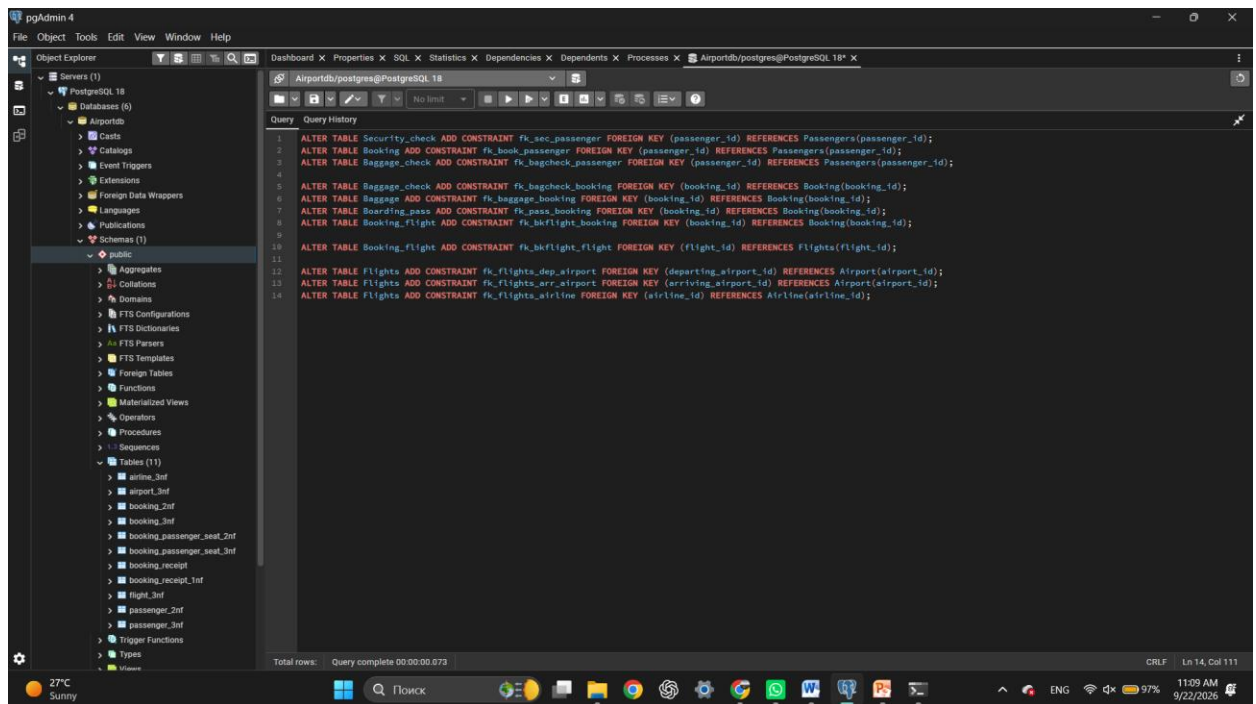
Creating all tables with PRIMARY KEY and NOT NULL constraints



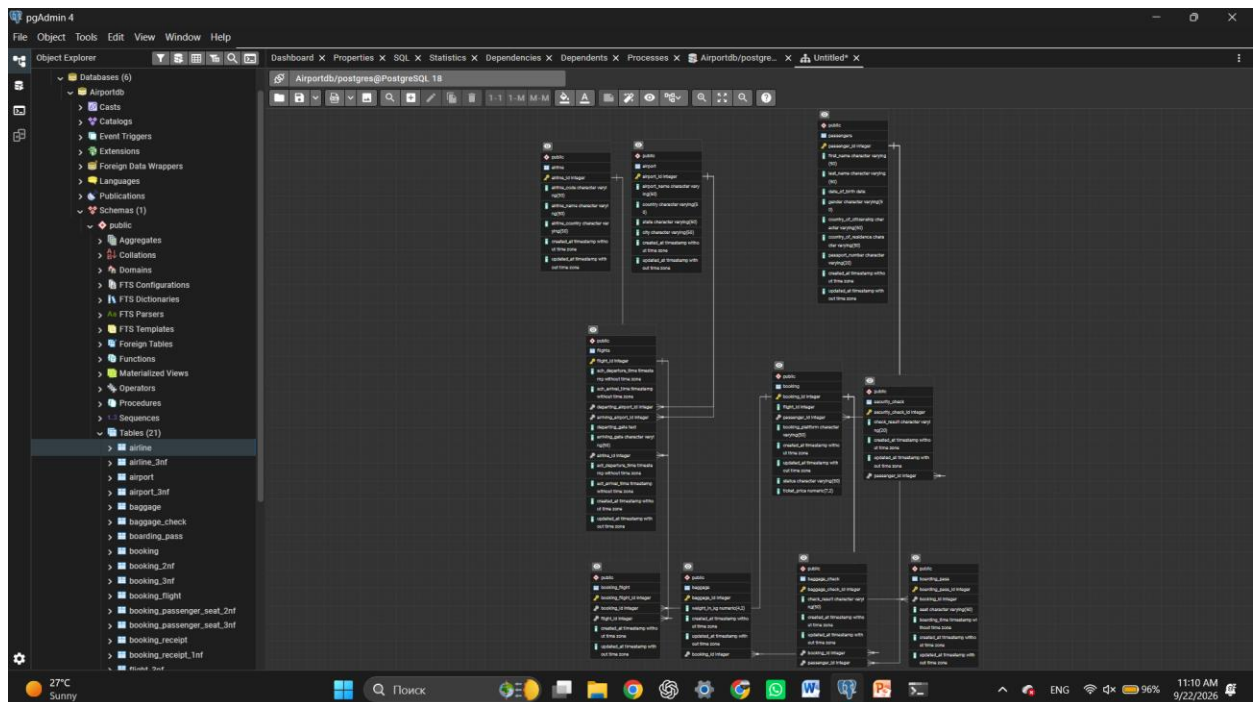
Modification of tables



Adding foreign keys (Relationships)



ERD:



Check whether the Flights table from the DDL section is in 3NF

Answer:

1. Current Primary Key (PK): `flight_id`
2. Functional Dependencies (FDs): `flight_id` → `sch_departure_time`, `sch_arrival_time`, `departing_airport_id`, `arriving_airport_id`, `departing_gate`, `arriving_gate`, `airline_id`, `act_departure_time`, `act_arrival_time`, `created_at`, `updated_at`
3. 3NF Analysis & Conclusion:

Is the table in 3NF? Yes, the Flights table is in 3NF

Explanation:

All non-key attributes depend directly on the primary key (`flight_id`).

The attributes `departing_airport_id`, `arriving_airport_id`, and `airline_id` are Foreign Keys (FKs) that reference other tables (Airport and Airline).

There are no transitive dependencies between non-key attributes. Nonkey attributes (like `departing_gate` or `sch_departure_time`) do not determine each other or depend on the foreign key IDs within this table.

```

64 );
65
66 CREATE TABLE Flights (
67     flight_id INT PRIMARY KEY,
68     sch_departure_time TIMESTAMP NOT NULL,
69     sch_arrival_time TIMESTAMP NOT NULL,
70     departing_airport_id INT NOT NULL,
71     arriving_airport_id INT NOT NULL,
72     departing_gate VARCHAR(50) NOT NULL,
73     arriving_gate VARCHAR(50) NOT NULL,
74     airline_id INT NOT NULL,
75     act_departure_time TIMESTAMP NOT NULL,
76     act_arrival_time TIMESTAMP NOT NULL,
77     created_at TIMESTAMP NOT NULL,
78     updated_at TIMESTAMP NOT NULL
79 );
80

```

